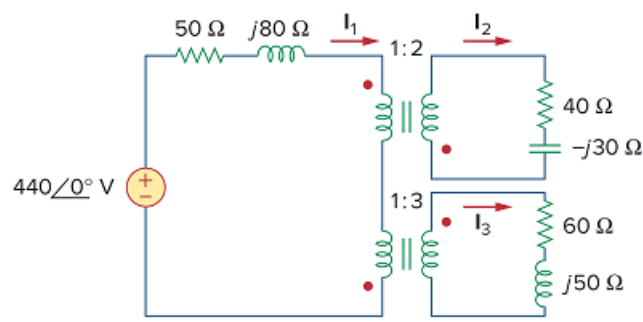


Problem

Determine I_1 , I_2 , and I_3 in the ideal transformer circuit of Fig. 13.145 using *PSpice* or *MultiSim*.

Figure 13.145



Step-by-step solution

Step 1 of 2

The schematic is shown below. We set total pts = 1, start frequency = 0.1592, and end frequency = 0.1592. After simulation, the output file includes,

Frequency	$I\ M(V_Print1)$	$I\ P(V_Print1)$
$1.592\ E-0.1$	$4.028\ E+00$	$-5.288\ E+0.1$

Frequency	$I\ M(V_Print2)$	$I\ P(V_Print2)$
$1.592\ E-0.1$	$2.019\ E+00$	$-5.211\ E+0.1$

Frequency	$I\ M(V_Print1)$	$I\ P(V_Print1)$
$1.592\ E-0.1$	$4.028\ E+00$	$-5.288\ E+0.1$

i.e.,

$$I_1 = 4.028 \angle -52.38^\circ\ A$$

$$I_2 = 2.019 \angle -52.11^\circ\ A$$

$$I_3 = 1.338 \angle -52.2^\circ\ A$$

Step 2 of 2

